Value of OSS Adoption for Grid Operators

Methodology:

- Grid operator’s choice between (1) proprietary and (2) OSS solutions
- Derived from game-theoretic discrete choice framework

Focal metric: Net benefit of OSS adoption measured in monetary terms

- Linear combination of value sources and weights (modular)
- Avoided costs, reduced risk, increased flexibility, downstream value

Key features:

- Defined over fixed operating period and known load
- Multi-stakeholder perspective: grid operator, regulator, downstream consumer
- Easily accommodates counterfactual analysis and uncertainty
- Customizable through flexible parameterization → highly extensible
- Open source!

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Timeline:

Year	Month	Phase
2026	2	Begin model development
	3	Model structure - feedback and refinement
	5	Model parameterization - feedback and refinement
	7	Report development
	9	Report publication and model release

Components of Benefit to Cost Analysis

Net benefit-to-cost is calculated as a **weighted** sum of

- **Benefit** components, **strategic Value (V)** and **Social benefits (S)**
- **Cost** components, **Total Cost of ownership (TCO)** and **Risk (R).**

Factor	Name	Type	Description
V	Strategic value	Benefit	Organization values, incl., Improvements to operational **efficiency** - reduced time-to-value i.e., **speed**, **innovations**, **Flexibility** (competitive procurement leverage), Multi-vendor **interoperability**/interchangeability, grid asset **utilization**
S	Social benefits	Benefit	Increased **Reliability/Resilience**; environmental **sustainability**, and social **equity**, consumer/prosumer services- improved economic opportunity,
TCO	Total cost of ownership	Cost	**CAPEX**: New specification design changes, qualifications, integration & commissioning **OPEX**: Operation & maintenance, upgrades, and support services, End-of-life
R	Risk	Cost	Service **reliability** risks, **regulatory** compliance, cyber **security**, and vendor **lock-in** risks

$$B_o = W_V \Delta V + W_s \Delta S - W_O \Delta TCO - W_R \Delta R$$

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Time	Topic	Lead

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Content Heavy Variants

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